REGION 2

Set up all six orders of integration for

$$\iiint\limits_{R_2} f(x,y,z)\,dV$$

where R_2 is the region bounded by the planes

$$z = 1 - x^2,\ z = 0,\ y = 1 - x,\ y = 0, \text{and } x = 0$$

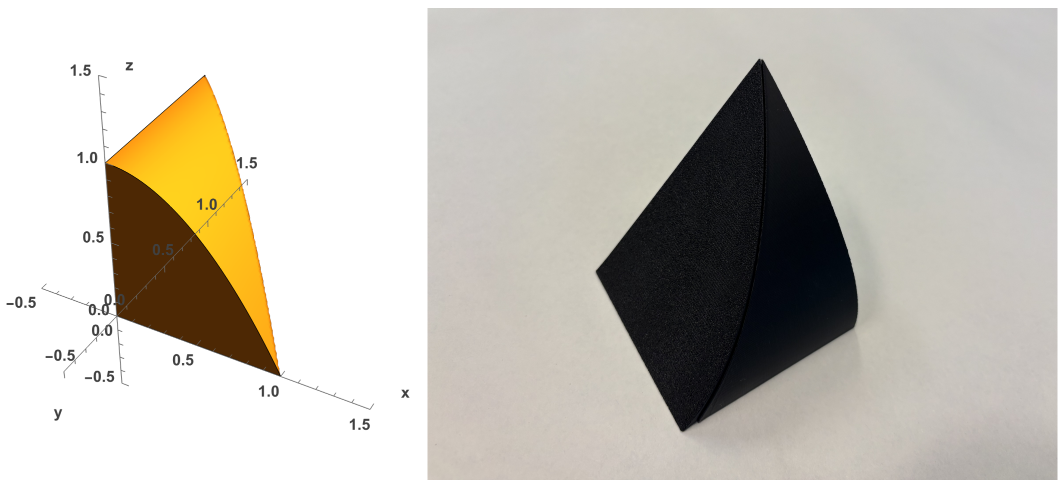
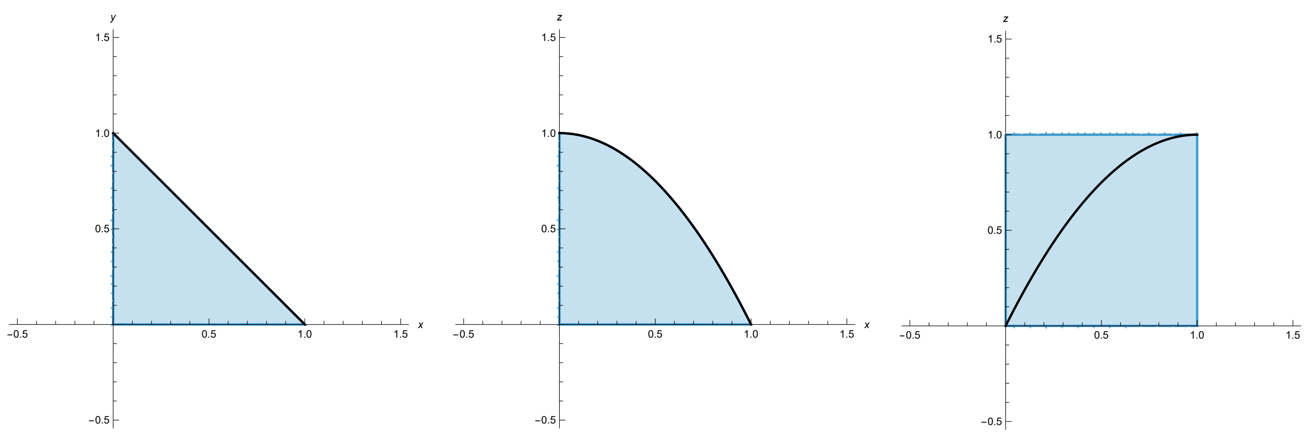


FIGURE 1. Computer-modeled and 3D-printed views of R_2.



(A) xy-projection. (B) xz-projection. (C) yz-projection.

FIGURE 2. Projections of R_2 into the three coordinate planes.

The following triple integrals represent integrating over the region using the xy-projection:

$$\int_0^1 \int_0^{1-y} \int_0^{1-x^2} f(x,y,z)\ dz\,dx\,dy$$

$$\int_0^1 \int_0^{1-x} \int_0^{1-x^2} f(x,y,z)\ dz\,dy\,dx$$

The following triple integrals represent integrating over the region using the xz-projection:

$$\int_0^1 \int_0^{\sqrt{1-z}} \int_0^{1-x} f(x,y,z)\ dy\,dx\,dz$$

$$\int_0^1 \int_0^{1-x^2} \int_0^{1-x} f(x,y,z)\ dy\,dz\,dx$$

The following triple integrals represent integrating over the region using the yz-projection:

$$\int_0^1 \int_0^{1-\sqrt{1-z}} \int_0^{\sqrt{1-z}} f(x,y,z)\ dx\,dy\,dz \ + \ \int_0^1 \int_{1-\sqrt{1-z}}^1 \int_0^{1-y} f(x,y,z)\ dx\,dy\,dz$$

$$\int_0^1 \int_{1-(1-y)^2}^1 \int_0^{\sqrt{1-z}} f(x,y,z)\ dx\,dz\,dy \ + \ \int_0^1 \int_0^{1-(1-y)^2} \int_0^{1-y} f(x,y,z)\ dx\,dz\,dy$$

All of these triple integrals evaluate to $\frac{5}{12}$ when $f(x,y,z) = 1$.